



# Briefing my colleagues

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# The Data

- The is for Goodnature traps.
- Traps has collected both qualitative and quantitative data sets.

## Qualitative data sets:

- Activity Type: Strike, Synced, and ect.
- Kill Type: Rat, Mouse, Stoat, and ect.

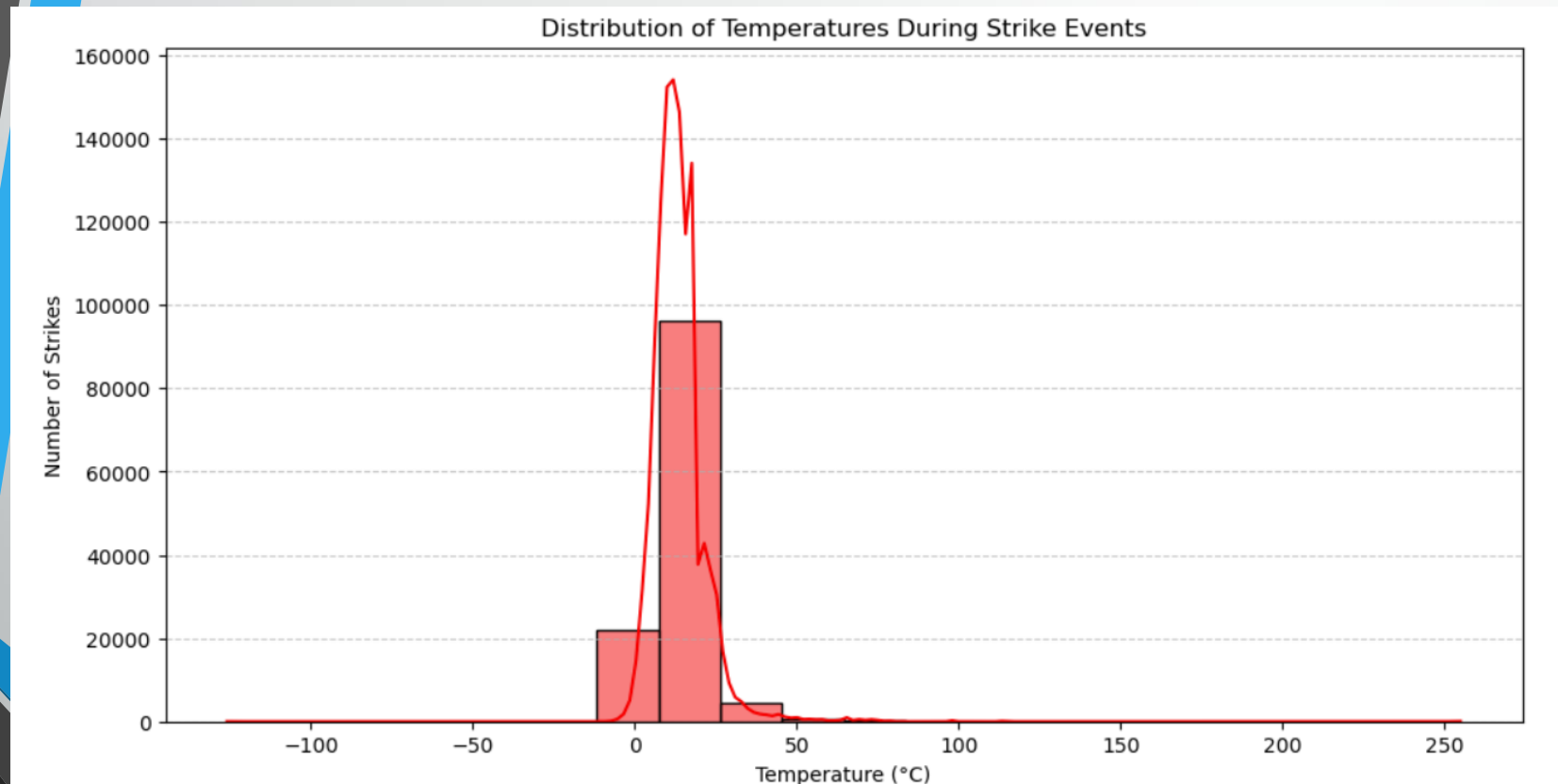
## Quantitative data sets:

- When: Year-Month-Day Hour:Minutes:Seconds +oo
- Strike time: " "
- Temperature: Celsius
- Latitude and Longitude: Coordinates system

# The Motivation

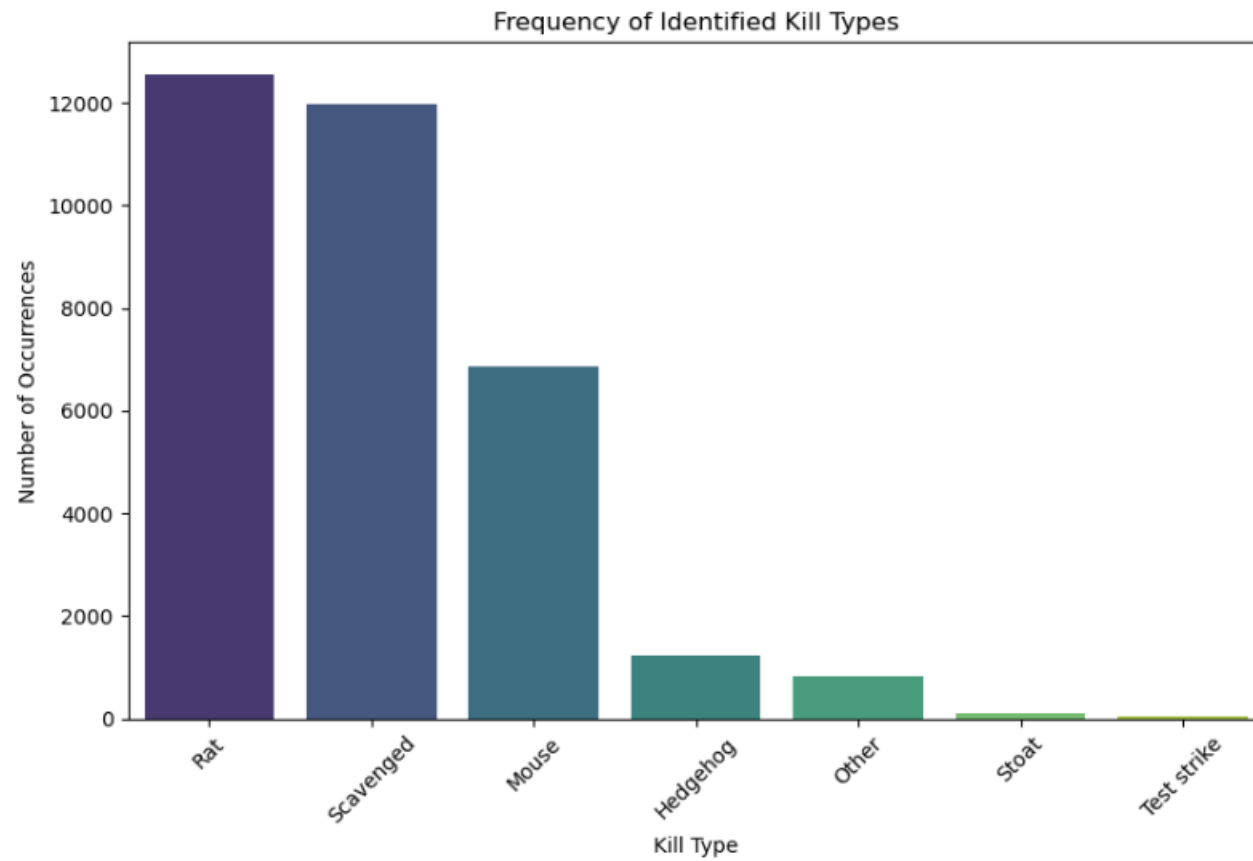
- Is the product achieving its purpose in reducing the population of invasive species in New Zealand?
- Does Temperature affect the rate of strikes (are there seasonal strikes)?
- Are the traps targeting invasive species?
- Is the location of the strike significant?

# What I've done so far



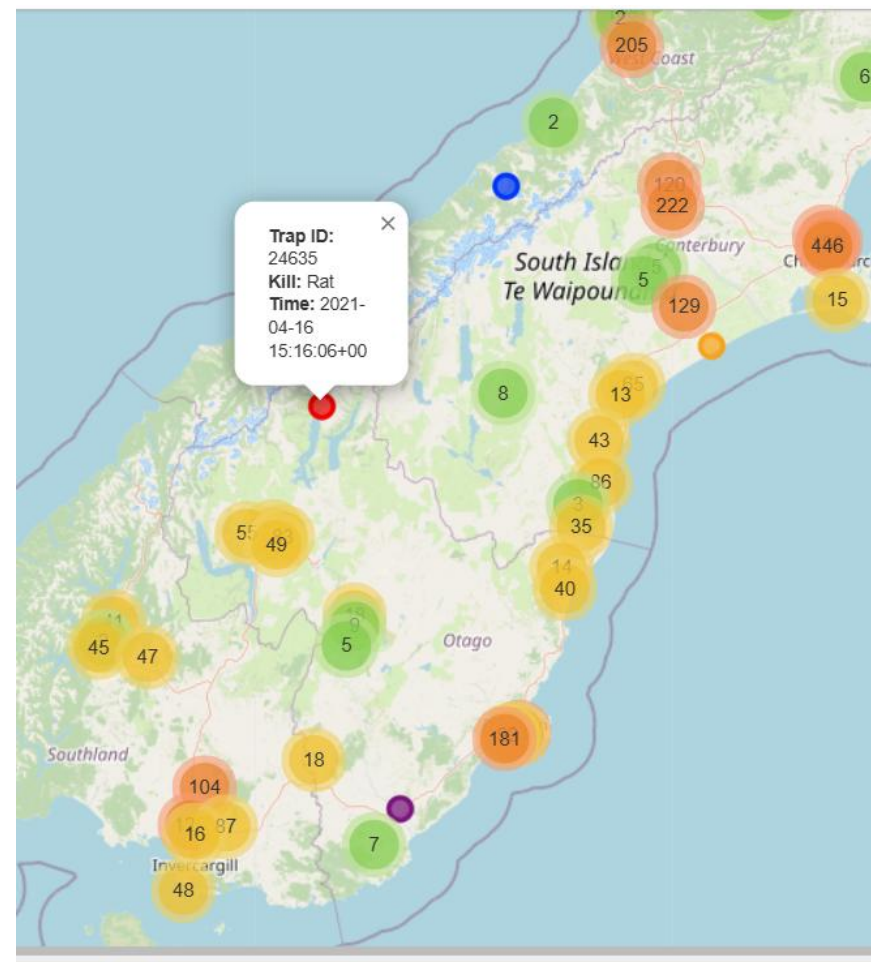
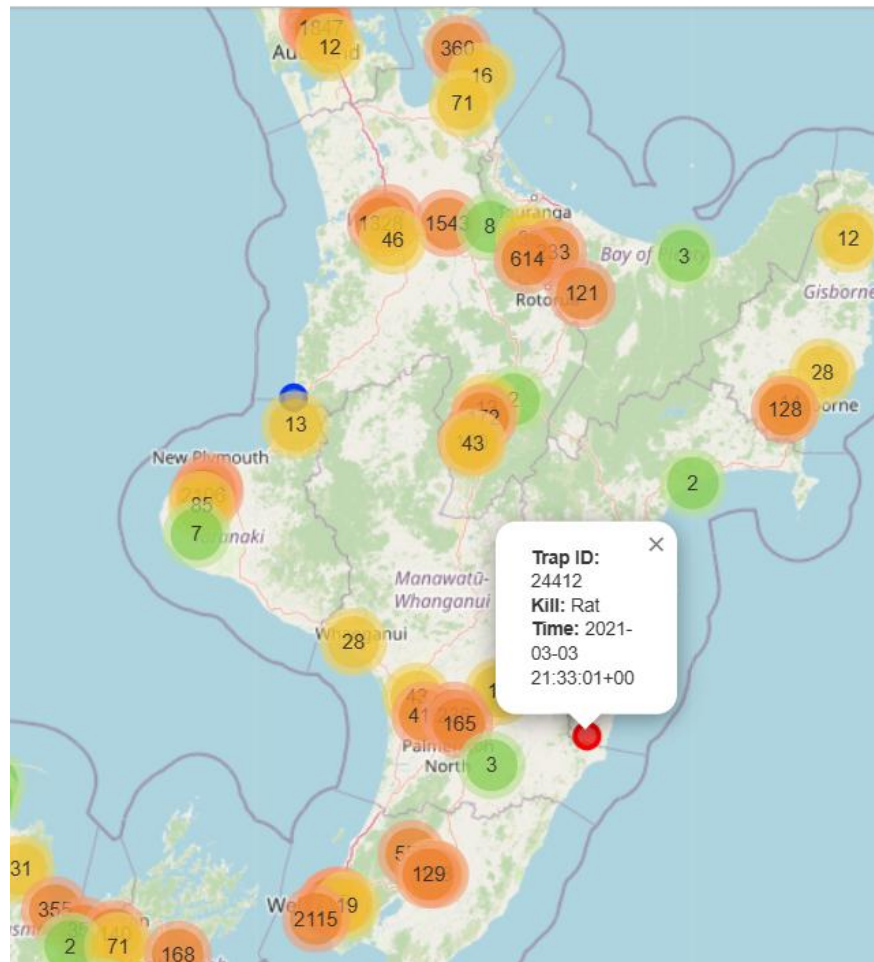
Temperature stats for strike events:

|       |               |
|-------|---------------|
| count | 123356.000000 |
| mean  | 13.672906     |
| std   | 8.408834      |
| min   | -126.000000   |
| 25%   | 9.000000      |
| 50%   | 13.000000     |
| 75%   | 17.000000     |
| max   | 255.000000    |



--- Kill Type Statistics (Excluding Missing Data) ---

|             | Total Count | Percentage (%) |
|-------------|-------------|----------------|
| killType    |             |                |
| Rat         | 12548       | 37.369706      |
| Scavenged   | 11966       | 35.636429      |
| Mouse       | 6861        | 20.433022      |
| Hedgehog    | 1225        | 3.648222       |
| Other       | 838         | 2.495682       |
| Stoat       | 90          | 0.268033       |
| Test strike | 50          | 0.148907       |





## Modelling the distribution of stoats on New Zealand offshore islands

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**Abstract:** Establishing the factors that correlate with the distribution of invasive species on islands provides a means of evaluating invasion potential and pathways, and enables the invasion risk for specific areas to be predicted. These island risk profiles then provide a basis for conservation managers to prioritise conservation areas, and to create appropriate monitoring regimes. In New Zealand, stoats (*Mustela erminea*) are invasive predators responsible for numerous local extinctions of birds, and they present a significant invasion risk for offshore islands. Here we examine the distribution of stoats on New Zealand's offshore islands in order to establish what factors are correlated with their distribution, and predict which islands display a high invasion risk. Data on the distribution of stoats were compiled, along with the characteristics of New Zealand's islands and their settlement history. The distribution of stoats on these islands was regressed on island characteristics

## Phylogeography of Invasive Rats in New Zealand

James C. Russell<sup>1\*</sup>, Judith H. Robins<sup>2</sup> and Rachel M. Fewster<sup>3</sup><sup>1</sup>School of Biological Sciences, University of Auckland, Auckland, New Zealand; <sup>2</sup>Department of Anthropology, University of Auckland, Auckland, New Zealand; <sup>3</sup>Department of Statistics, University of Auckland, Auckland, New Zealand

Two species of invasive rats (*Rattus norvegicus* and *R. rattus*) arrived in New Zealand with Europeans in the mid to late eighteenth and nineteenth century respectively. They rapidly spread across the main islands of New Zealand and its offshore islands, displacing the historically introduced *R. exulans*. Today both species are widespread although the distribution of the sub-dominant *R. norvegicus* is patchy. Tissue samples were obtained from 425 *R. rattus* and 130 *R. norvegicus* across the New Zealand

## Modelling the dynamics and control of stoats in New Zealand forests

SCIENCE FOR CONSERVATION 252

N.D. Barlow and M.C. Barron



Management of Biological Invasions (2022) Volume 13, Issue 3: 466–478

Special Issue: Effectiveness of Goodnature A24 self-resetting rat traps for invasive rodent control  
Guest editors: Aaron B. Shiels and Christopher A. Lepczyk

## Editorial

## An introduction to a special issue and review of the effectiveness of Goodnature A24 self-resetting rat traps

Aaron B. Shiels<sup>1\*</sup>, Tyler Bogardus<sup>2</sup>, Lisa H. Crampton<sup>3</sup>, Markus Gronwald<sup>4</sup>, Abby M. Kreuser<sup>1</sup>, Roger A. Baldwin<sup>1</sup> and Christopher A. Lepczyk<sup>1</sup><sup>1</sup>USDA, APHIS, Wildlife Services, National Wildlife Research Center, 4101 LaPorte Avenue, Fort Collins, CO, 80521 USA<sup>2</sup>USDA, APHIS, Wildlife Services, Honolulu, HI, 96721 USA, and University of Hawaii, Office of the Vice President for Research and Innovation, in support of the U.S. Army Garrison, Natural Resources Program on Oahu, Schofield Barracks, HI, 96857, USA<sup>3</sup>Kaui Forest Bird Recovery Project, Pacific Cooperative Studies Unit, 3751 Hanapepe Rd, Hanapepe, HI, 96716, USA<sup>4</sup>Forest Protection Division, Ministry of Natural Resources and Environment, P.O. Box 108, Wellington, New Zealand

# Other resources



Thank you